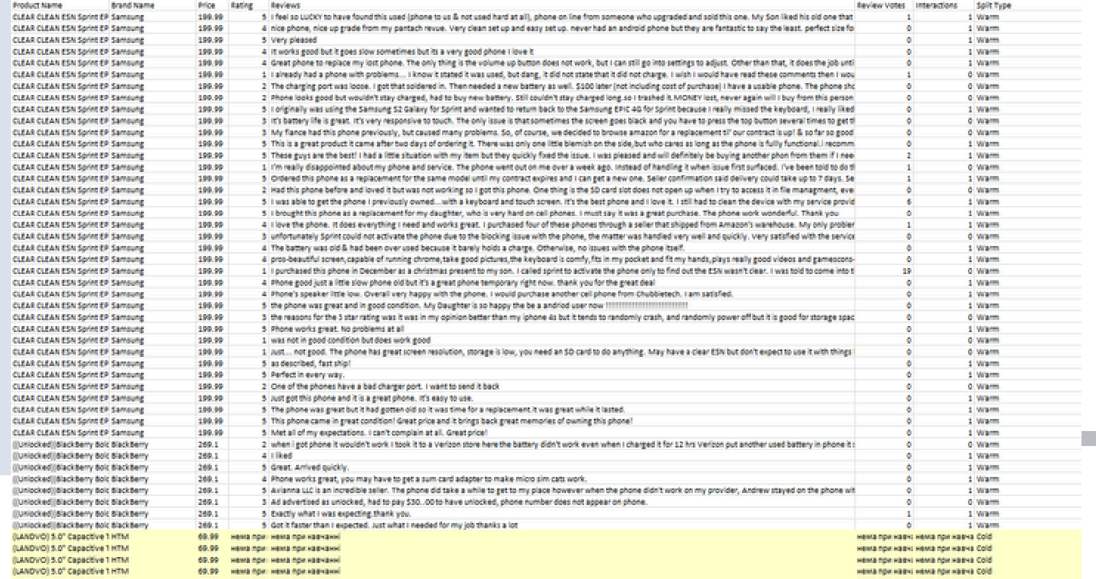
Датасети даних

- Для дослідження обрано реальний датасет телефонної продукції від Amazon. Він містить і поведінкову історію, і характеристики пристроїв
- Телефони з Amazon обрано саме тому, що вони мають характеристику ″Бренд″ та ″Модель″, чого немає в інших датасетах, як в фільмах (MovieLens)
- Для рекомендацій є важливий факт задоволеності покупкою: якщо рейтинг товару більше 4 зірок, їх переведено в активні зв’язки, а оцінки з 1,2 або 3 зірками відкинуто



Product Name	Brand Name	Price	Rating	Reviews	Review Votes	Interactions	Split Type
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	I feel so LUCKY to have found this used (phone to us & not used hard at all), phone on line from someone who upgraded and sold this one. My Son liked his old one that	1	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	4	nice phone, nice up grade from my pantach revue. Very clean set up and easy set up. never had an android phone but they are fantastic to say the least. perfect size fo	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	Very pleased	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	4	It works good but it goes slow sometimes but its a very good phone I love it	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	4	Great phone to replace my lost phone. The only thing is the volume up button does not work, but I can still go into settings to adjust. Other than that, it does the job unti	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	1	I already had a phone with problems... I know it stated it was used, but dang, it did not state that it did not charge. I wish I would have read these comments then I wou	1	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	2	The charging port was loose. I got that soldered in. Then needed a new battery as well. $100 later (not including cost of purchase) I have a usable phone. The phone sho	0	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	2	Phone looks good but wouldn't stay charged, had to buy new battery. Still couldn't stay charged long.so I trashed it.MONEY lost, never again will I buy from this person.	0	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	I originally was using the Samsung S2 Galaxy for Sprint and wanted to return back to the Samsung EPIC 4G for Sprint because I really missed the keyboard, I really liked	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	3	It's battery life is great. It's very responsive to touch. The only issue is that sometimes the screen goes black and you have to press the top button several times to get t	0	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	My fiance had this phone previously, but caused many problems. So, of course, we decided to browse amazon for a replacement til' our contract is up! & so far so good	0	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	This is a great product it came after two days of ordering it. There was only one little blemish on the side,but who cares as long as the phone is fully functional.I recomm	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	These guys are the best! I had a little situation with my item but they quickly fixed the issue. I was pleased and will definitely be buying another phon from them if I nee	2	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	1	I'm really disappointed about my phone and service. The phone went out on me over a week ago. Instead of handling it when issue first surfaced. I've been told to do th	1	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	Ordered this phone as a replacement for the same model until my contract expires and I can get a new one. Seller confirmation said delivery could take up to 7 days. Se	1	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	2	Had this phone before and loved it but was not working so I got this phone. One thing is the SD card slot does not open up when I try to access it in file managment, eve	0	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	I was able to get the phone I previously owned... with a keyboard and touch screen. It's the best phone and I love it. I still had to clean the device with my service provid	6	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	I brought this phone as a replacement for my daughter, who is very hard on cell phones. I must say it was a great purchase. The phone work wonderful. Thank you	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	4	I love the phone. It does everything I need and works great. I purchased four of these phones through a seller that shipped from Amazon's warehouse. My only probler	1	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	3	unfortunately Sprint could not activate the phone due to the blocking issue with the phone, the matter was handled very well and quickly. Very satisfied with the servic	0	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	4	The battery was old & had been over used because it barely holds a charge. Otherwise, no issues with the phone itself.	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	4	pros-beautiful screen,capable of running chrome,take good pictures,the keyboard is comfy,fits in my pocket and fit my hands,plays really good videos and gamescons-	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	1	I purchased this phone in December as a christmas present to my son. I called sprint to activate the phone only to find out the ESN wasn't clear. I was told to come into t	19	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	4	Phone good just a little slow phone old but it's a great phone temporary right now. thank you for the great deal	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	4	Phone's speaker little low. Overall very happy with the phone. I would purchase another cell phone from Chubbietech. I am satisfied.	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	the phone was great and in good condition. My Daughter is so happy the be a andriod user now !!!!!!!!!!!!!!!!!!!!!!!!!!	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	3	the reasons for the 3 star rating was it was in my opinion better than my iphone 4s but it tends to randomly crash, and randomly power off but it is good for storage spac	0	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	Phone works great. No problems at all	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	1	was not in good condition but does work good	0	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	1	Just... not good. The phone has great screen resolution, storage is low, you need an SD card to do anything. May have a clear ESN but don't expect to use it with things	0	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	as described, fast ship!	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	Perfect in every way.	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	2	One of the phones have a bad charger port. I want to send it back	0	0	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	Just got this phone and it is a great phone. It's easy to use.	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	The phone was great but it had gotten old so it was time for a replacement.It was great while it lasted.	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	This phone came in great condition! Great price and it brings back great memories of owning this phone!	0	1	Warm
CLEAR CLEAN ESN Sprint EP	Samsung	199.99	5	Met all of my expectations. I can't complain at all. Great price!	0	1	Warm
((Unlocked))BlackBerry Bol	BlackBerry	269.1	2	when I got phone it wouldn't work I took it to a Verizon store here the battery didn't work even when I charged it for 12 hrs Verizon put another used battery in phone it	0	0	Warm
((Unlocked))BlackBerry Bol	BlackBerry	269.1	4	I liked	0	1	Warm
((Unlocked))BlackBerry Bol	BlackBerry	269.1	5	Great. Arrived quickly.	0	1	Warm
((Unlocked))BlackBerry Bol	BlackBerry	269.1	4	Phone works great, you may have to get a sum card adapter to make micro sim cats work.	0	1	Warm
((Unlocked))BlackBerry Bol	BlackBerry	269.1	5	Avianna LLC is an incredible seller. The phone did take a while to get to my place however when the phone didn't work on my provider, Andrew stayed on the phone wit	0	1	Warm
((Unlocked))BlackBerry Bol	BlackBerry	269.1	3	Ad advertised as unlocked, had to pay $30..00 to have unlocked, phone number does not appear on phone.	0	0	Warm
((Unlocked))BlackBerry Bol	BlackBerry	269.1	5	Exactly what I was expecting.thank you.	1	1	Warm
((Unlocked))BlackBerry Bol	BlackBerry	269.1	5	Got it faster than I expected. Just what I needed for my job thanks a lot	0	1	Warm
(LANDVO) 5.0" Capactive 1	HTM	69.99	нема при	нема при навчанні	нема при навч	нема при навча	Cold
(LANDVO) 5.0" Capactive 1	HTM	69.99	нема при	нема при навчанні	нема при навч	нема при навча	Cold
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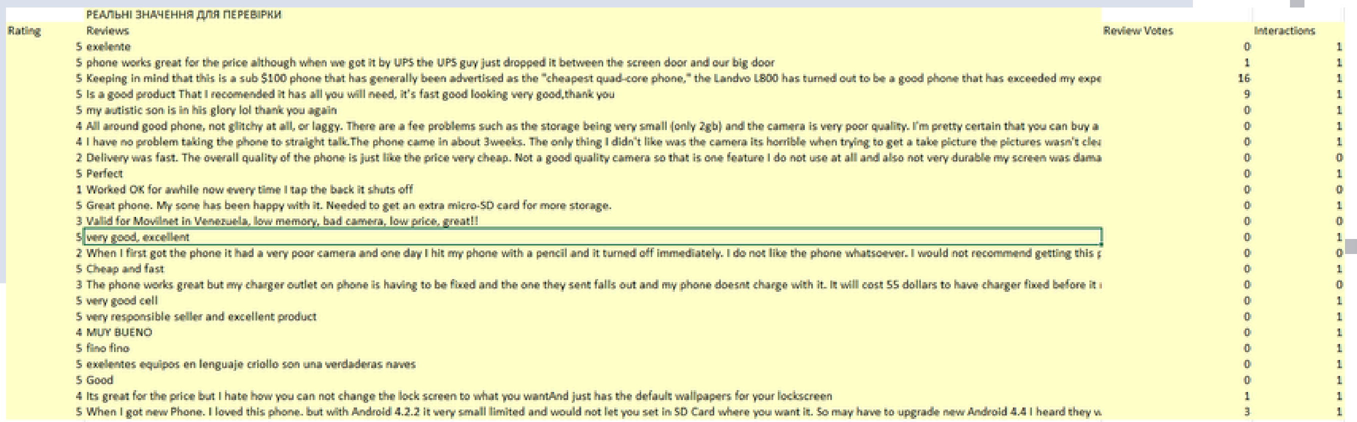
(LANDVO) 5.0" Capactive 1	HTM	69.99	нема при	нема при навчанні	нема при навч	нема при навча	Cold
(LANDVO) 5.0" Capactive 1	HTM	69.99	нема при	нема при навчанні	нема при навч	нема при навча	Cold

Симуляція cold start

- Із 4 моделей телефонів (Samsung, BlackBerry, HTM, Jethro) 3 моделі призначено ‘’теплими’’ - на цих значеннях модель вчиться будувати зв’язки
- З навчального графа спеціально видалено ребра з моделі HTM. Для моделі у цього товару немає історії продаж.
- У графі тільки залишено характеристики пристрою: його тип, ціна та бренд. Завдання GNN поєднати новий товар з потрібними покупцями.
- Усі справжні взаємодії користувачів з телефоном відведено в інший файл (true_results). Після навчання дані моделі звірюються з прихованими фактами.



РЕАЛЬНІ ЗНАЧЕННЯ ДЛЯ ПЕРЕВІРКИ

Rating	Reviews	Review Votes	Interactions
5	exelente	0	1
5	phone works great for the price although when we got it by UPS the UPS guy just dropped it between the screen door and our big door	1	1
5	Keeping in mind that this is a sub $100 phone that has generally been advertised as the "cheapest quad-core phone," the Landvo L800 has turned out to be a good phone that has exceeded my expe	16	1
5	Is a good product That I recomended it has all you will need, it's fast good looking very good,thank you	9	1
5	my autistic son is in his glory lol thank you again	0	1
4	All around good phone, not glitchy at all, or laggy. There are a fee problems such as the storage being very small (only 2gb) and the camera is very poor quality. I'm pretty certain that you can buy a	0	1
4	I have no problem taking the phone to straight talk.The phone came in about 3weeks. The only thing I didn't like was the camera its horrible when trying to get a take picture the pictures wasn't cle	0	1
2	Delivery was fast. The overall quality of the phone is just like the price very cheap. Not a good quality camera so that is one feature I do not use at all and also not very durable my screen was dama	0	0
5	Perfect	0	1
1	Worked OK for awhile now every time I tap the back it shuts off	0	0
5	Great phone. My sone has been happy with it. Needed to get an extra micro-SD card for more storage.	0	1
3	Valid for Movilnet in Venezuela, low memory, bad camera, low price, great!!	0	0
5	very good, excellent	0	1
2	When I first got the phone it had a very poor camera and one day I hit my phone with a pencil and it turned off immediately. I do not like the phone whatsoever. I would not recommend getting this	0	0
5	Cheap and fast	0	1
3	The phone works great but my charger outlet on phone is having to be fixed and the one they sent falls out and my phone doesnt charge with it. It will cost 55 dollars to have charger fixed before it	0	0
5	very good cell	0	1
5	very responsible seller and excellent product	0	1
4	MUY BUENO	0	1
5	fino fino	0	1
5	exelentes equipos en lenguaje criollo son una verdaderas naves	0	1
5	Good	0	1
4	Its great for the price but I hate how you can not change the lock screen to what you wantAnd just has the default wallpapers for your lockscreen	1	1
5	When I got new Phone. I loved this phone. but with Android 4.2.2 it very small limited and would not let you set in SD Card where you want it. So may have to upgrade new Android 4.4 I heard they w	3	1